Given Network Information
IPv4: 172.20.0.0/21
Host Information

Sales	280 Hosts	/23
Marketing	90 Hosts	/25
Administration	28 Hosts	/27
IT	8 Hosts	/28

Step 1: Subnet the given subnet and assign the 1st subnet to Sales department while wasting the fewest addresses.
Step 2: From the available subnets that you created in Step 1, subnet further the 2nd subnet as per the host requirement of Marketing department.
Step 3: From the subnets that you created in Step 2, assign the 1st subnet to Marketing department and subnet further the 3rd subnet as per the requirement of Administration department.
Step 4: From the subnets that you created in Step 3, assign the 1st subnet to Administration department and subnet further the2nd subnet as per the requirement of IT department and assign the 2nd subnet to IT Department.

Show step-by-step solution and answer the following:
a. Network Address, Broadcast Address, Subnet Mask of each assigned subnet to Sales, Marketing, Administration, and IT department
b. Find the 255th and 280th Host IP of Sales Department.
c. Find the 50th Host IP of Marketing Department
d. Find the 8th Host IP of IT Department

Solution:
Given Network
172.20.00000 000.00000000/21

Step 1: Subnet the given subnet and assign the 1st subnet to Sales department while wasting the fewest addresses.
/21 → /23

172.20.00000	00	0.00000000	Assigned to Sales
	01		/23 -- /25
	10		
	11		

Step 2: From the available subnets that you created in Step 1, subnet further the 2nd subnet as per the host requirement of Marketing department.
/23 → /25

172.20.00000	01	0.0	0000000	Assigned to Marketing
		0.1		
		1.0		/25 -- /27
		1.1		

Step 3: From the subnets that you created in Step 2, assign the 1st subnet to Marketing department and subnet further the 3rd subnet as per the requirement of Administration department.
/25 -- /27

172.20.00000	01	1.0	00	00000	Assigned to Administration
			01		/27 -- /28
			10		
			11		

Step 4: From the subnets that you created in Step 3, assign the 1st subnet to Administration department and subnet further the 2nd subnet as per the requirement of IT department and assign the 2nd subnet to IT Department.

/27 -- /28

172.20.00000 01 1.0 01 0 0000

1 Assigned to IT

Summary

Sales

172.20.0000000 0.00000000/23 NA: 172.20.0.0/23 SM: 255.255.254.0

0.11111111 #255: 172.20.0.255

1.00011000 #280: 172.20.1.24

1.11111111 BA: 172.20.1.255

Marketing

172.20.00000010.0 0000000/25 NA: 172.20.2.0/25 SM: 255.255.255.128

1111111 BA: 172.20.2.127

Administration

172.20.00000011.000 00000/27 NA: 172.20.3.0/27 SM: 255.255.255.224

11111 BA: 172.20.3.31

IT

172.20.00000011.0011 0000/28 NA: 172.20.3.48/28 SM: 255.255.255.240

1111 BA: 172.20.3.63